

MINOR PROJECT – SUPERVISED LEARNING

Project Requirements: Students must complete both parts in a single notebook.

Notebook Structure:

- **Part A:** Regression (Medical Cost Prediction)
- **Part B:** Classification (Movie Rating Prediction)

PART A – REGRESSION

Medical Cost Prediction

Objective: Build a Linear Regression model to predict medical insurance charges.

SECTION 1 – Data Understanding (Mandatory)

1. **Load the dataset and display:**
 - Shape
 - Column names
 - Data types
 - First 5 rows
2. **Check:**
 - Missing values
 - Duplicate rows
3. **Perform EDA:**
 - Distribution of charges
 - Scatter plot: age vs charges
 - Boxplot: smoker vs charges
 - Correlation heatmap
4. **Write 3 observations from EDA.**

SECTION 2 – Data Preprocessing

5. Encode categorical variables properly.
6. Separate features (X) and target (y).
7. Perform train-test split (80-20).
8. Apply feature scaling (if required).
9. Explain why scaling is / isn't required.



SECTION 3 – Model Building

10. Train a Linear Regression model.
11. **Display:**
 - Intercept
 - Coefficients
12. **Write the regression equation.**

SECTION 4 – Model Evaluation

13. Predict on test data.
14. **Calculate:**
 - MAE
 - MSE
 - RMSE
 - R2 Score
15. **Plot:**
 - Actual vs Predicted
 - Residual plot
16. **Interpret R2 score.**

SECTION 5 – Business Insights

17. Which variable impacts charges the most?
18. How much more do smokers pay?
19. Is BMI statistically impactful?
20. Can this model be used in production? Why or why not?



PART B – CLASSIFICATION

Movie Rating Prediction

Objective: Build a classification model to predict whether a movie is a Hit or Flop.

Define:

- **Hit** = rating ≥ 7
- **Flop** = rating < 7

SECTION 1 – Data Understanding

21. **Load dataset and display:**
 - Shape
 - Columns
 - Data types
22. Check missing values.
23. Create binary target column: **Hit**
24. Show class distribution.
25. Plot rating distribution.
 - Write 3 observations.

SECTION 2 – Data Preprocessing

26. Drop irrelevant columns.
27. Encode categorical variables.
28. Separate X and y.
29. Train-test split (80-20).
30. Scale if required.

SECTION 3 – Model Building

31. Train Logistic Regression model.
32. Display model coefficients.
33. Explain what coefficients mean.

SECTION 4 – Evaluation

34. Predict test data.
35. **Calculate:**
 - Accuracy
 - Precision
 - Recall
 - F1 Score



36. Plot Confusion Matrix.

37. Explain:

- Why accuracy alone is not enough?
- Which metric matters more here?

SECTION 5 – Business Interpretation

38. What factors influence movie success?

39. Is the dataset balanced?

40. Would you trust this model for production? Why?

Submission Guidelines

- Single Jupyter Notebook (.ipynb)
- Proper section headings
- Clean code
- Clear interpretation after every major step
- No copied notebooks
- No skipping evaluation metrics
- Both models compulsory



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